ASEN 3802 Lab 1 Part 3: Structural Mechanics (Truss Lab)

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Smead Aerospace CU Boulder ASEN 3802: Aerospace Laboratory

Part 3 Task 1

Undeformed and Deformed Truss Geometry (FEM Results)

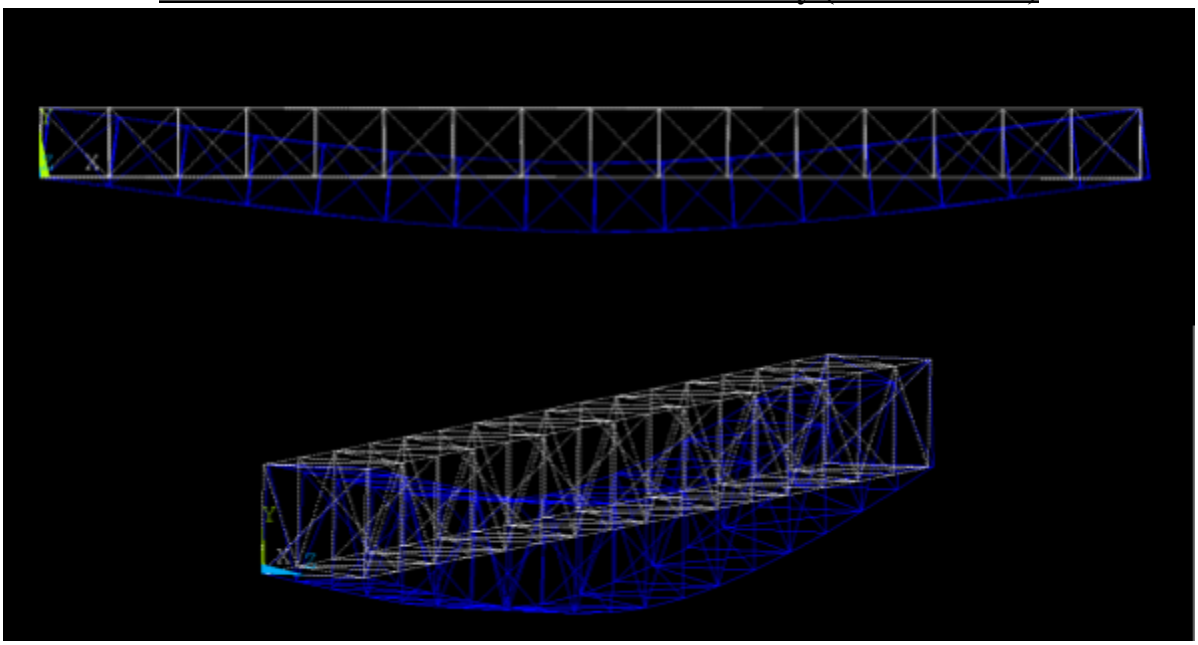

Fig (1): Figure 1 illustrates the truss in an iso and side view, both of the deformed model and the model under no loading. The generated shape is from ANSYS,

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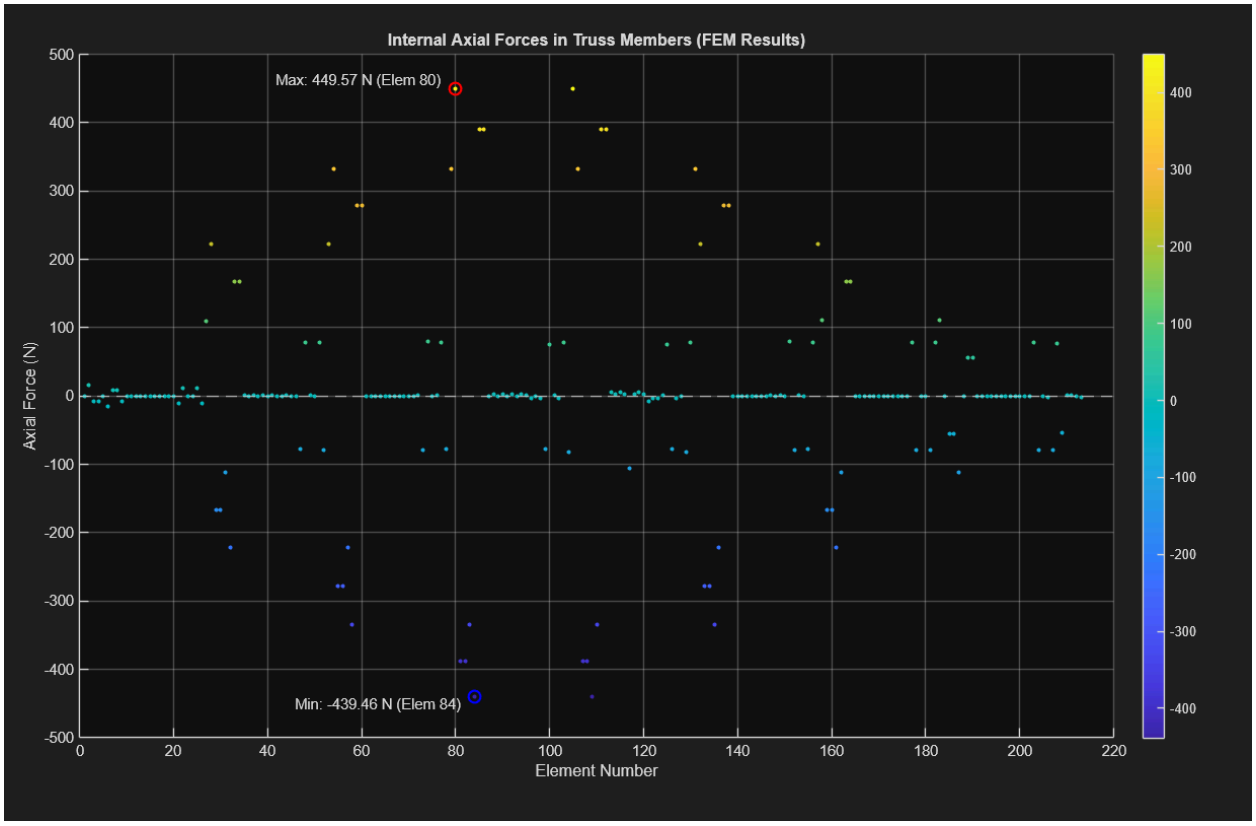

Fig (2): The contour plot depicts the loads in the truss. The Max load was 449.57N, and the min was -439.46N; the sign indicates whether it was in compression or tension; a negative sign indicates compression, and the opposite for tension.

Discussion:

The ANSYS Model Figure 1 shows undeformed and deformed truss geometry derived from the FEM simulation. The maximum displacement happens around midspan, and the deformation pattern shows a smooth downward deflection. This reaction is in line with the behavior of classical beams and is typical of a simply supported structure under a centered vertical load. Both the imposed boundary restrictions and the symmetric loading condition are reflected in the symmetry of the deformation.

The ANSYS model for the deformed shape of the truss shows that the bending behavior dominates the structural response. Members along the upper portion of the truss where the loads being applied primarily experience compressive forces, while members along the lower portion experience tensile forces. This distribution is mechanically consistent, as bending produces tensile and compressive stress leading to compression in the upper region of the structure and tension in the lower region.

The ANSYS Model Figure 2 presents the internal axial force distribution across the truss members. The scatter plot does not directly identify the physical location of individual elements, but shows a symmetrical pattern of force distribution along the truss members. This symmetry mirrors the displacement behavior observed in the deformed geometry. Since the applied load

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and support conditions are symmetric, the internal force response is expected to display similar symmetry. The presence of this pattern provides qualitative confirmation of the mechanical consistency of the FEM model.

The axial force results show both tensile and compressive forces, demonstrating the load-sharing behavior inherent to truss structures. The maximum tensile force was observed to be 449.57 N, while the maximum compressive force reached −439.46 N. Members near midspan carry the largest axial forces, which is physically reasonable because the bending moment is maximized at midspan under centered loading case. The magnitude of maximum tensile and compressive force does not exactly mirror itself because there are assumed to be non-linear effects across the beam that ANSYS does not take into account.

Diagonal members exhibit varying force magnitudes, reflecting their role in transferring shear forces and redistributing loads throughout the structure. Overall, the FEM results demonstrate structurally consistent behavior, including symmetric deformation and an axial force distribution characteristic of bending-dominated response.

Part 3 Task 2

Comparison of Analytical, FEM, and Experimental Results

	Experimental	Equiv. Beam Model	ANSYS (FEM)
Internal Axial Froce in longeron strut near midspan [N]	282.15	444.57	389.91
% Error		37.53 %	27.64 %
Verticle Displacement [mm]	1.473	1.711	1.856
% Error		13.91%	20.64 %
Reaction forces [N]	L support: 116.8 R support: 113.2	L support: 111.2 R support: 111.2	L support: 111.2 R support: 111.2
% Error		L: 4.79 % R: 1.77 %	L: 4.79 % R: 1.77 %

Fig (3)

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Axial Force Calculation from Experimental Data

Experimental Axial Force:

at 0 lbf load:

- $F3D_0 = -85.763$ lbf

at 50 lbf (222.4 N) load:

- $F3D_{50} = -22.334$ lbf

- $\Delta F3D = (-22.334) - (-85.763)$

$= 63.429$ lbf

$63.429\,(4.4482) = \boxed{282.15\ N}$

Fig (4)

FEM Truss Model with Element Numbering

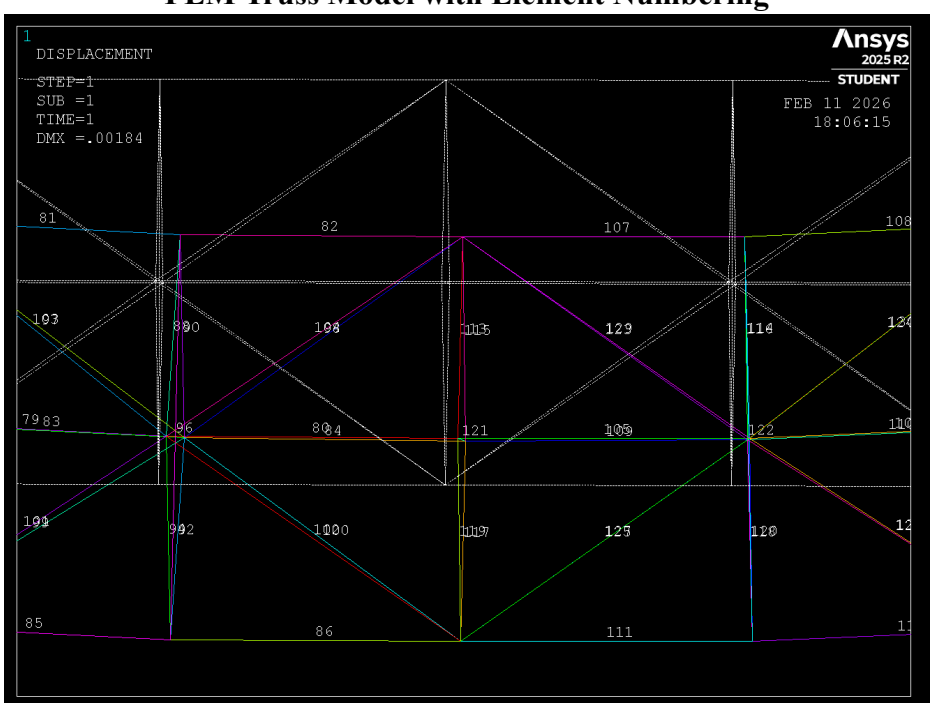

Fig (5)

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Axial Force Calculation for the Equiv. Beam Model at the Midspan

Equivalent Beam Model Axial Force at center load

- $P = 222.4\ N$
- $A = 0.061359\ in^2 = 3.958E-5\ m^2$
- $c = 0.25/2 = 0.125\ m$
- $I = 5.947\ in^4 \left(0.0254 \frac{m}{in}\right)^4 \approx 2.475E-6\ m^4$

- $\sigma_{LC} = \frac{M_{mid}\, c}{I} \Rightarrow F_{beam} = \sigma_{LC} A = \frac{M_{mid}\, c}{I} A$

- $F_{beam} = \frac{(222.4)(0.125)}{2.475E-6}(3.958E-5)$

$= \boxed{444.57\ N}$

Fig (6)

Axial Force Calculation using Experimental Data

Experimental Axial Force:

at 0 lbf load:

- $F3D_0 = -85.763\ lbf$

at 50 lbf (222.4 N) load:

- $F3D_{50} = -22.334\ lbf$

- $\Delta F3D = (-22.334) - (-85.763)$

$= 63.429\ lbf$

$63.429\ (4.4482) = \boxed{282.15\ N}$

Fig (7)

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Displacement Calculation for the Equiv. Beam Model

- $\Sigma I = 5.947(in)^4 = 2.475\ E\text{-}6\ m^4$
- $L = 16(0.25) = 4.0\ m$
- $P = 222.4\ N$
- $E = 70E9\ Pa$

$$v = \frac{1}{EI}\left(\frac{PL^3}{48}\right)$$

$$= \frac{(22.4)(4)^3}{(70E9)(2.475E\text{-}6)(48)}$$

$$= 0.001711\ m$$

$$= \boxed{1.711\ mm}$$

Fig (8)

Calculation of Reaction Forces Using Experimental Data

Experimental Reaction Forces:

- $A_y = F0 + F1$
- $C_y = F2$

- $F0_{50} = 11.762\ lbf$
- $F1_{50} = 14.497\ lbf$
- $F2 = 25.449\ lbf = \boxed{113.2\ N}$
- $A_y = 11.762\ lbf + 14.497\ lbf$

$$= 26.259\ lbf$$

$$= \boxed{116.8\ N}$$

Fig (9)

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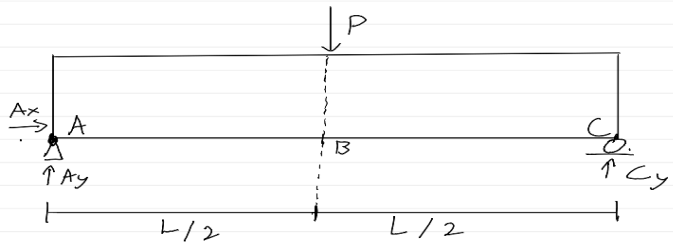
Calculation of Reaction Forces For the Equiv. Beam Model

Equivalent Beam Model Reaction Forces:



• $P = 222.4\ N$ • $L = 4.0\ m$

$$\Sigma F_y = A_y + C_y - P = 0$$
$$= A_y + C_y = 222.4\ N$$

• $$\Sigma M_A = P(L/2) - C_y(L) = 0$$
$$= 222.4(2) - C_y(4) = 0$$
$$= C_y = \frac{444.8}{4} = \boxed{111.2\ N}$$

• $$A_y + 111.2\ N = 222.4\ N$$
$$= A_y = \boxed{111.2\ N}$$

Fig (10)

Discussion:

Axial Forces:

The equivalent beam model predicts the largest axial force among the three methodologies. This result is expected because the beam idealization assumes that bending resistance is carried entirely by the beams while neglecting the contribution of diagonal members. As a result, the bending stress and corresponding axial force in the lower beam are over-estimated. The reaction force calculation is given as: A_y (Equivalent Beam model) > A_y (Experimental Data).

The FEM model yields a lower axial force compared to the beam prediction. This behavior reflects the more realistic load distribution provided by the truss model, where diagonal members participate in carrying the applied load. Consequently, the axial force demand on individual beams is reduced relative to the simplified beam assumption.

For the FEM results, the axial force from element 86 **(Seen in Fig 5)** was selected for comparison. This element corresponds to a lower beam strut located at the midspan, consistent with the experimental configuration. The inline load cell measures the internal force across a longitudinal beam member near midspan, making this element the appropriate reference for comparison. Although FEM provides the maximum axial force across all members, using the

force from the element corresponding to the load cell location ensures consistency between numerical and experimental measurements.

The experimental axial force is lower than both analytical predictions. This discrepancy is attributed to several factors inherent to the physical truss build-up or setup, including but not limited to structural loading, sensor calibration offsets, and minor variations in load transfer through joints and fittings. The truss might be assumed to be not perfectly horizontally aligned when load is being applied leading to discrepancy in the value. Some loads might be transferred through minor infractions or loosen joints and bolts. Thus, load might not be distributed proportionally across each truss member. Additionally, real boundary conditions deviate from idealized supports, influencing internal force distribution.

Displacement:

The FEM model predicts the largest vertical displacement, while the equivalent beam model yields a slightly smaller value. This difference arises from modeling assumptions. The equivalent beam model represents a highly simplified stiffness idealization that ignores diagonal members and joint flexibility, considering more stiffness or rigid truss model.

The FEM simulation incorporates the full truss geometry, allowing deformation mechanisms associated with axial member deformation to develop. This produces a more flexible structural response and therefore a larger predicted displacement. The FEM simulation for truss geometry can be considered a real-world scenario about how trusses behave when load is being applied.

The experimental displacement is slightly smaller than both predictions. This behavior is consistent with real structural effects not captured in idealized models, such as local joints stiffness, fixture constraints, and measurement uncertainty including sensor offset position could also influence measured deflection.

Reaction Forces:

Both the equivalent beam model and FEM simulation predict symmetric reaction forces. This result follows directly from static equilibrium and geometric symmetry under centered loading. The agreement between these methodologies proves that force balance across truss members is accurately represented.

The experimental reaction forces show slight asymmetry and a marginally higher total reaction force. These deviations are expected in a physical setup and are attributed to load cell offsets, truss fixture imbalance, joints or bolts infraction, support friction, and minor geometric imperfections. Unlike the idealized analytical and FEM models, the real structure experiences non-uniform load transfer and boundary effects.

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Part 3 Task 3

Material Stiffness:

The axial stiffness of LINK180 elements is proportional to Young's modulus(E). To model the potential effects of joint free-play, Young's modulus was reduced to 90% and 80% of its nominal value. These variations represent mild and moderate stiffness reductions and provide a reasonable sensitivity range

CASE I & II: Impact on Axial Force, Vertical Displacement and Reaction Forces due to Change in Young's Modulus

	Baseline	Case 1 (90% Eo)	% Change from Baseline	Case 2 (80% Eo)	% Change from Baseline
Young's Modulus value [Pa]	7.00E+10	6.30E+10	10 %	5.60E+10	20 %
Internal Axial Froce in ELEMENT 86 [N]	389.91	389.61	- 7.69E-2 %	389.9	- 2.56E-3 %
Verticle Displacement [mm]	-1.856	-2.062	11.1 %	-2.3	23.9 %
Reaction forces [N]	L support: 111.2 R support: 111.2	L support: 111.2 R support: 111.2	0 %	L support: 111.2 R support: 111.2	0 %

Fig (11)

Discussion:

Reducing Young's modulus produced a noticeable increase in vertical displacement resulting in a near proportional change between both; while the axial force in element 86 remained nearly unchanged. This behavior is consistent with linear structural mechanics. Since displacement is inversely proportional to stiffness, reductions in Young's modulus directly increased the structural deflection.

In contrast, the internal axial force exhibited minimal sensitivity to stiffness variation. This result is mechanically reasonable because member forces in a nearly statically determinate truss are governed primarily by equilibrium rather than stiffness distribution.

Reaction forces remained unchanged across all cases, confirming that global force balance is independent of material stiffness assumptions.

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Free-play:

Free-play within the inline load cell may introduce additional compliance, reducing the effective axial stiffness of the member containing the sensor. Since LINK180 elements behave as axial members with stiffness governed by $k = \frac{EA}{L}$, this effect can be modeled by reducing Young's modulus for the specific element corresponding to the load cell location.

To investigate this uncertainty, the stiffness of element 86 was reduced while maintaining nominal properties for all other elements. Two stiffness reduction cases were considered: 90% and 80% of the nominal Young's modulus. These variations simulate mild and moderate compliance effects associated with sensor free-play.

CASE I & II: Impact on Axial Force, Vertical Displacement and Reaction Forces due to Change in Young's Modulus (Element 86)

	Baseline	Case 1 (Elem 86 @ 90% Stiffness)	% Change from Baseline	Case 2 (Elem 86 @ 80% Stiffness)	% Change from Baseline
Young's Modulus (Element 86 only) [Pa]	7.00E+10	6.30E+10	10 %	5.60E+10	20 %
Internal Axial Froce in ELEMENT 86 [N]	389.91	384.95	- 1.23 %	379.28	- 2.80 %
Verticle Displacement [mm]	-1.856	-1.862	0.32 %	-1.87	0.75 %
Reaction forces [N]	L support: 111.2 R support: 111.2	L support: 111.2 R support: 111.2	0 %	L support: 111.2 R support: 111.2	0 %

Fig (12)

Discussion:

Reducing the stiffness of element 86 produced a small but consistent decrease in the axial force carried by that member. When element 86 stiffness was reduced to 90% of baseline, the axial force decreased by approximately −1.23%, and when reduced to 80% of baseline, the axial force decreased by approximately −2.80%. This trend is mechanically consistent: making a single member more compliant causes it to attract slightly less load as forces redistribute to surrounding members.

The global vertical displacement changed only slightly, with changes on the order of less than 1% relative to baseline. This indicates that modifying the stiffness of one member has minimal influence on the overall structural stiffness and global deflection, which is expected because the stiffness reduction is localized.

Reaction forces remained unchanged, confirming that global equilibrium is not affected by localized stiffness modifications. Overall, the results suggest that load cell free-play can reduce the measured axial force in the instrumented member and therefore may contribute to the experimental discrepancy. However, at the 10–20% stiffness reduction level, the magnitude of the force reduction is limited, implying that additional factors (such as support friction, load

redistribution, or sensor calibration effects) are also likely influencing the measured axial force.

Friction:

Idealized boundary conditions assume frictionless supports. In the physical experiment, however, friction at the roller supports may introduce resistance to longitudinal motion. This resistance can alter the internal force distribution while having only a limited effect on the global vertical deformation.

To investigate this uncertainty, support friction was approximated by applying small horizontal forces at the roller nodes opposing longitudinal displacement. Two friction sensitivity cases were evaluated using total longitudinal forces of 10 N and 20 N. These values represent realistic friction levels relative to the applied vertical load.

CASE I & II: Impact on Axial Force, Vertical Displacement and Reaction Forces due to Applied Frictional Force

	Baseline	Case 1: Total of 10 N of Simulated Friction Force	Change from Baseline	Case 2: Total of 20 N of Simulated Friction Force	Change from Baseline
Combined Friction Force at Rollers [N]	0.00E+00	1.00E+01	10 N	2.00E+01	20 N
Internal Axial Froce in ELEMENT 86 [N]	389.91	384.81	- 1.33 %	380.01	- 2.61 %
Verticle Displacement [mm]	-1.856	-1.841	- 0.81 %	-1.826	- 1.64 %
Reaction forces [N]	L support: 111.2 R support: 111.2	L support: 111.2 R support: 111.2	0 %	L support: 111.2 R support: 111.2	0 %

Fig (13)

Discussion:

Applying simulated friction at the roller supports produced small but consistent changes in both internal axial forces and vertical displacement. The axial force in element 86 decreased by −1.33% for a 10 N friction case and by −2.61% for a 20 N friction case. This trend is mechanically consistent: friction introduces additional constraint forces at the supports, which modifies the internal load path and redistributes axial forces among members.

Vertical displacement decreased in magnitude by approximately 0.81% and 1.64% for the 10 N and 20 N cases, respectively. This indicates that friction slightly increases the effective restraint in the system, producing a small stiffening effect in the global response. Vertical reaction forces remained unchanged, confirming that global vertical equilibrium is governed primarily by the applied vertical load, while friction influences internal force distribution and horizontal equilibrium.

Overall, the results suggest that support friction can contribute to reducing the predicted axial force in the load-cell member, though the effect remains modest at the 10–20 N friction levels considered. Therefore, friction may partially explain experimental discrepancies but is unlikely

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to be the sole cause.

Manufacturing Imperfections:

Manufacturing imperfections may reduce the cross-sectional dimensions of truss members, which lowers axial stiffness through the relation $k = \frac{EA}{L}$. To model potential fabrication thinning, the cross-sectional area of all LINK180 elements was reduced by 5% and 10% relative to the nominal value while keeping material properties unchanged.

CASE I & II: Impact on Axial Force, Vertical Displacement and Reaction Forces due to Reduced Cross-sectional Area

	Baseline	Case 1: Reduced Cross-sectional area by 5%	% Change from Baseline	Case 2: Reduced Cross-sectional area by 10%	% Change from Baseline%
Cross-sectional Area [m^2]	0.00	0.00	- 5 %	0.00	- 10 %
Internal Axial Froce in ELEMENT 86 [N]	389.91	389.61	-0.077 %	389.61	-0.077 %
Verticle Displacement [mm]	-1.86	-1.95	4.62 %	-2.06	10.75 %
Reaction forces [N]	L support: 111.2 R support: 111.2	L support: 111.2 R support: 111.2	0 %	L support: 111.2 R support: 111.2	0 %

Fig (14)

Discussion:

Reducing the cross-sectional area increased the magnitude of vertical displacement while producing negligible changes in the axial force carried by element 86. The displacement magnitude increased by approximately 4.6% for a 5% area reduction and 10.8% for a 10% area reduction, which is mechanically consistent because axial stiffness scales with area.

In contrast, the axial force in element 86 remained essentially unchanged (about −0.08% in both cases). This indicates that the internal force distribution is governed primarily by equilibrium and truss geometry rather than uniform stiffness scaling. Since all members were reduced proportionally, the overall load in each member was preserved, resulting in similar member forces while allowing greater deformation due to reduced overall stiffness. Reaction forces were unchanged, as expected from global equilibrium.

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Bending moment and stiffness:

The baseline FEM model employed LINK180 elements, which assume purely axial member behavior and ideal pin-jointed connections with no bending stiffness. In practical structures, however, members possess bending rigidity and joints may transmit moments due to connection details and friction effects.

To evaluate the influence of this modeling assumption, the truss was alternatively modeled using BEAM188 elements, which incorporate bending stiffness and rotational degrees of freedom. The beam formulation allows members to deform through flexure in addition to axial deformation. The results obtained from the BEAM188 model were compared directly to the baseline LINK180 simulation using identical geometry, loading conditions, and boundary constraints.

Comparison of LINK180 and BEAM188 Models

	Link Simulation	Beam Simulation	% Change from Link to Beam
Internal Axial Froce in ELEMENT 86 [N]	389.91	389.61	0 %
Verticle Displacement [mm]	-1.86	-2.32	24.73 %
Reaction forces [N]	L support: 111.2 R support: 111.2	L support: 111.2 R support: 111.2	0 %

Fig (15)

Discussion:

Switching from LINK180 to BEAM188 produced no changes in the axial force carried by element 86, while significantly affecting the vertical displacement response. The axial force remained essentially unchanged, indicating that internal force distribution is governed primarily by equilibrium and truss geometry rather than element formulation.

In contrast, the vertical displacement magnitude increased by approximately 24.73%. This behavior is mechanically consistent, as the beam formulation introduces bending flexibility and rotational degrees of freedom that are absent in the bar-element model. The beam elements allow members to deform through flexure in addition to axial stretching, resulting in increased global deformation when bending stiffness is relatively small.

Reaction forces remained unchanged across both models, confirming that global equilibrium is independent of structural idealization. Overall, the comparison demonstrates that displacement predictions are sensitive to element formulation and bending stiffness assumptions, while axial force predictions remain largely governed by equilibrium constraints.

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Contribution Report - Structural Mechanics

Ritik Sarraf: I've contributed to Part 3 Discussion portion.

Quinn Jochim: N/A

Anton Pylypchenko: I did all of the ANSYS simulations, written computations, and Task 3 discussions.

Harroop Sooch: Description for first two figures.

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Appendix:

Ansys Code:

```
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SECDATA,3.958637244E-5
SECNUM,0
MP,EX,1,7.0E10
MP,PRXY,1,0.3

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n,18,0,0.25,0
n,35,0,0.25,0.25
n,52,0,0,0.25
et,1,link180
ngen,17,1,1,52,17,0.25,0,0,1

E,1,2
E,2,3
E,18,19
E,19,20
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E,36,37
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E,52,2
E,2,54
E,18,2
E,2,20
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```
Egen,8,2,all
E,17,34
E,34,51
E,51,68
E,68,17
E,51,17

/SOLU
ANTYPE,0
D,1,ALL,0
D,52,ALL,0
D,17,UY,0
D,17,UZ,0
D,68,UY,0
D,68,UZ,0
F,26,FY,-111.2
F,43,FY,-111.2

SOLVE
FINISH

/POST1
SET,LAST
PLDISP,2
PRNSOL,U,Y
ETABLE,RES1,SMISC,1
PRETAB,RES1
PRRSOL
```

Internal Forces Data:

ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]	ELEMENT	FORCE [N]
1	-0.16209	42	-0.19112	83	-334.27	124	0.25671	165	0.11385	247	-7.7941	288	7.67E-02	329	-9.41E-02	370	15.732
2	15.732	43	-0.44149	84	-439.46	125	74.537	166	-7.72E-02	248	-7.79E+00	289	0.14302	330	-0.10814	371	-7.7941
3	-7.7941	44	0.27029	85	390.25	126	-78.42	167	0.11385	249	0	290	0	331	11.131	372	-7.7941
4	-7.7941	45	0.21517	86	390.25	127	-4.0392	168	-7.72E-02	250	-1.59E+01	291	0.14302	332	-11.333	373	0
5	0	46	5.51E-02	87	-0.39692	128	0.15642	169	0.11385	251	8.2435	292	-0.10845	333	-7.9213	374	-15.897
6	-15.897	47	-78.369	88	2.7461	129	-82.724	170	-7.72E-02	252	8.24E+00	293	-0.20226	334	1.69E-02	375	8.2435
7	8.2435	48	78.639	89	-0.39692	130	78.841	171	0.11385	253	-7.8708	294	-0.10845	335	1.69E-02	376	8.2435
8	8.2435	49	0.31546	90	2.7461	131	332.36	172	-7.72E-02	254	1.43E-01	295	-9.38E-02	336	0.00E+00	377	-7.8708
9	-7.8708	50	-4.52E-02	91	-0.39692	132	222.05	173	-0.16101	255	7.67E-02	296	-1.13E+01	337	-2.39E-02	378	0.14302
10	0.14302	51	78.891	92	2.7461	133	-277.65	174	0.10924	256	0.14302	297	11.133	338	-0.16209	379	7.67E-02

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11	7.67E-02	52	-78.621	93	-0.39692	134	-277.65	175	1.43E-02	257	7.67E-02	298	-9.41E-02	339	1.57E+01	380	0.14302
12	0.14302	53	221.76	94	2.7461	135	-334.41	176	9.49E-02	258	1.43E-01	299	-0.10814	340	-7.7941	381	7.67E-02
13	7.67E-02	54	332.22	95	0.56133	136	-222.47	177	78.691	259	0	300	11.131	341	-7.7941	382	0.14302
14	0.14302	55	-277.87	96	-3.8836	137	278.7	178	-78.581	260	0.14302	301	-11.333	342	0	383	0
15	0	56	-277.87	97	0.15841	138	278.7	179	0.11463	261	-0.10845	302	-7.9213	343	-15.897	384	0.14302
16	0.14302	57	-222.19	98	-4.042	139	-0.397	180	-5.38E-03	262	-2.02E-01	303	1.69E-02	344	8.24E+00	385	-0.10845
17	-0.10845	58	-334.27	99	-78.426	140	-9.15E-02	181	-78.57	263	-0.10845	304	1.69E-02	345	8.24E+00	386	-0.20226
18	-0.20226	59	278.91	100	74.542	141	-0.397	182	78.679	264	-9.38E-02	305	0.00E+00	346	-7.8708	387	-0.10845
19	-0.10845	60	278.91	101	0.25869	142	-9.15E-02	183	110.94	265	-11.335	306	-2.39E-02	347	1.43E-01	388	-9.38E-02
20	-9.38E-02	61	0.14769	102	-4.1423	143	-0.397	184	-6.01E-13	266	1.11E+01	307	-0.16209	348	7.67E-02	389	-11.335
21	-11.335	62	-9.82E-02	103	78.835	144	-9.15E-02	185	-55.331	267	-9.41E-02	308	1.57E+01	349	0.14302	390	11.133
22	11.133	63	0.14769	104	-82.718	145	-0.397	186	-55.331	268	-0.10814	309	-7.7941	350	7.67E-02	391	-9.41E-02
23	-9.41E-02	64	-9.82E-02	105	449.57	146	-9.15E-02	187	-111.32	269	11.131	310	-7.7941	351	0.14302	392	-0.10814
24	-0.10814	65	0.14769	106	332.36	147	0.56145	188	0	270	-11.333	311	0	352	0	393	11.131
25	11.131	66	-9.82E-02	107	-388.72	148	0.12939	189	55.94	271	-7.9213	312	-15.897	353	0.14302	394	-11.333
26	-11.333	67	0.14769	108	-388.72	149	0.30474	190	55.94	272	1.69E-02	313	8.24E+00	354	-0.10845	395	-7.9213
27	110.13	68	-9.82E-02	109	-439.46	150	-0.17535	191	2.44E-02	273	1.69E-02	314	8.24E+00	355	-0.20226	396	1.69E-02
28	221.76	69	-0.20887	110	-334.41	151	78.981	192	0.20975	274	0	315	-7.8708	356	-0.10845	397	1.69E-02
29	-167.06	70	0.13893	111	390.18	152	-78.852	193	2.44E-02	275	-2.39E-02	316	1.43E-01	357	-9.38E-02	398	0
30	-167.06	71	-0.26398	112	390.18	153	0.40503	194	0.20975	276	-0.16209	317	7.67E-02	358	-1.13E+01	399	-2.39E-02
31	-111.28	72	0.40292	113	5.7852	154	-0.27563	195	2.44E-02	277	1.57E+01	318	0.14302	359	11.133		
32	-222.19	73	-78.848	114	2.7455	155	-78.28	196	0.20975	278	-7.7941	319	7.67E-02	360	-9.41E-02		
33	167.46	74	78.987	115	5.7852	156	78.409	197	2.44E-02	279	-7.79E+00	320	0.14302	361	-0.10814		
34	167.46	75	-0.1637	116	2.7455	157	222.05	198	0.20975	280	0	321	0	362	11.131		
35	0.31218	76	0.30263	117	-105.41	158	110.94	199	-3.45E-02	281	-1.59E+01	322	0.14302	363	-11.333		
36	-0.19112	77	78.412	118	2.7455	159	-166.78	200	-0.29664	282	8.2435	323	-0.10845	364	-7.9213		
37	0.31218	78	-78.273	119	5.7852	160	-166.78	201	-0.12944	283	8.2435	324	-0.20226	365	1.69E-02		
38	-0.19112	79	332.22	120	2.7455	161	-222.47	202	-0.1672	284	-7.8708	325	-0.10845	366	1.69E-02		
39	0.31218	80	449.57	121	-8.1815	162	-111.32	203	78.547	285	0.14302	326	-9.38E-02	367	0.00E+00		
40	-0.19112	81	-388.79	122	-3.8828	163	167.03	204	-78.843	286	7.67E-02	327	-1.13E+01	368	-2.39E-02		
41	0.31218	82	-388.79	123	-4.1395	164	167.03	205	-2.92E-02	287	1.43E-01	328	11.133	369	-0.16209		

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[4] Harroop Sooch, ASEN 3802, CU Boulder

Matlab Code for Axial Forces Plot:

```
clc, clear, close all;
data = readmatrix('Truss_Int_Forces.csv');

% Remove text
data(1:2,:) = [];

% Store data into variables
elem = data(:,1);
force = data(:,2);

% Find max tension and compression
[maxForce, idxMax] = max(force);
[minForce, idxMin] = min(force);
elemMax = elem(idxMax);
elemMin = elem(idxMin);

% Plota
figure
scatter(elem, force, 8, force,"filled")
colorbar
hold on
yline(0, '--')

% Highlight max and min
plot(elemMax, maxForce, 'ro', 'MarkerSize', 8, 'LineWidth', 1.5)
plot(elemMin, minForce, 'bo', 'MarkerSize', 8, 'LineWidth', 1.5)
% Labels

xlabel('Element Number')
ylabel('Axial Force (N)')
title('Internal Axial Forces in Truss Members (FEM Results)')
grid on

% Annotate
text(elemMax - 80, maxForce, ...
   sprintf('  Max: %.2f N (Elem %d)', maxForce, elemMax), ...
   'VerticalAlignment','bottom')
text(elemMin - 80, minForce, ...
   sprintf('  Min: %.2f N (Elem %d)', minForce, elemMin), ...
   'VerticalAlignment','top')

% Print to command window
fprintf('Maximum axial force: %.2f N at element %d\n', maxForce, elemMax);
fprintf('Minimum axial force: %.2f N at element %d\n', minForce, elemMin);
```

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Ansys code for visualizing bar element numbering:

```
/prep7

SECTYPE,0,LINK,,LINK1
SECDATA,3.958637244E-5
SECNUM,0

MP,EX,1,7.0E10
MP,PRXY,1,0.3

n,1,0,0,0
n,18,0,0.25,0
n,35,0,0.25,0.25
n,52,0,0,0.25

et,1,link180

ngen,17,1,1,52,17,0.25,0,0,1

E,1,2
E,2,3
E,18,19
E,19,20
E,35,36
E,36,37
E,52,53
E,53,54
E,1,18
E,2,19
E,18,35
E,19,36
E,35,52
E,36,53
E,52,1
E,53,2
E,1,35
E,53,19
E,18,36
E,36,20
E,52,36
E,36,54
E,52,2
E,2,54
```

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[4] Harroop Sooch, ASEN 3802, CU Boulder

```
E,18,2
E,2,20

Egen,8,2,all

E,17,34
E,34,51
E,51,68
E,68,17
E,51,17

/SOLU
ANTYPE,0

D,1,ALL,0
D,52,ALL,0
D,17,UY,0
D,17,UZ,0
D,68,UY,0
D,68,UZ,0

F,26,FY,-111.2
F,43,FY,-111.2

SOLVE
FINISH

/POST1
SET,LAST

/PNUM,ELEM,1
EPLOT


PLDISP,2

PRNSOL,U,Y

ETABLE,RES1,SMISC,1
PRETAB,RES1

PRRSOL
```

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Task 3: Ansys code for Free-play Case:

```
/prep7

SECTYPE,0,LINK,,LINK1
SECDATA,3.958637244E-5
SECNUM,0

MP,EX,1,7.0E10
MP,PRXY,1,0.3

MP,EX,2,6.3E10
MP,PRXY,2,0.3

MP,EX,3,5.6E10
MP,PRXY,3,0.3

MP,PRXY,1,0.3

n,1,0,0,0
n,18,0,0.25,0
n,35,0,0.25,0.25
n,52,0,0,0.25

et,1,link180

ngen,17,1,1,52,17,0.25,0,0,1

E,1,2
E,2,3
E,18,19
E,19,20
E,35,36
E,36,37
E,52,53
E,53,54
E,1,18
E,2,19
E,18,35
E,19,36
E,35,52
E,36,53
E,52,1
E,53,2
```

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[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

```
E,1,35
E,53,19
E,18,36
E,36,20
E,52,36
E,36,54
E,52,2
E,2,54
E,18,2
E,2,20

Egen,8,2,all

E,17,34
E,34,51
E,51,68
E,68,17
E,51,17

EMODIF,86,MAT,2

/SOLU
ANTYPE,0

D,1,ALL,0
D,52,ALL,0
D,17,UY,0
D,17,UZ,0
D,68,UY,0
D,68,UZ,0

F,26,FY,-111.2
F,43,FY,-111.2

SOLVE
FINISH

/POST1
SET,LAST

/PNUM,ELEM,1
EPLOT
```

[1] Quinn Jochim, ASEN 3802, CU Boulder
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[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

```
PLDISP,2

PRNSOL,U,Y

ETABLE,RES1,SMISC,1
PRETAB,RES1

PRRSOL
```

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Task 3: Ansys code for Friction Cases:

```
/prep7

SECTYPE,0,LINK,,LINK1
SECDATA,3.958637244E-5
SECNUM,0

MP,EX,1,7.0E+10
MP,PRXY,1,0.3

n,1,0,0,0
n,18,0,0.25,0
n,35,0,0.25,0.25
n,52,0,0,0.25

et,1,link180

ngen,17,1,1,52,17,0.25,0,0,1

E,1,2
E,2,3
E,18,19
E,19,20
E,35,36
E,36,37
E,52,53
E,53,54
E,1,18
E,2,19
E,18,35
E,19,36
E,35,52
E,36,53
E,52,1
E,53,2
E,1,35
E,53,19
E,18,36
E,36,20
E,52,36
E,36,54
E,52,2
E,2,54
```

```
E,18,2
E,2,20

Egen,8,2,all

E,17,34
E,34,51
E,51,68
E,68,17
E,51,17

/SOLU
ANTYPE,0

D,1,ALL,0
D,52,ALL,0
D,17,UY,0
D,17,UZ,0
D,68,UY,0
D,68,UZ,0

F,26,FY,-111.2
F,43,FY,-111.2

F,17,FX,-10
F,68,FX,-10

SOLVE
FINISH

/POST1
SET,LAST

/PNUM,ELEM,1
EPLOT


PLDISP,2

PRNSOL,U,Y

ETABLE,RES1,SMISC,1
PRETAB,RES1
```

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PRRSOL

Bending Moment and stiffness:

```
/prep7

SECTYPE,1,BEAM,CTUBE,BEAM1
SECDATA,0.01,0.015
SECNUM,1

MP,EX,1,5.6E+10
MP,PRXY,1,0.3

n,1,0,0,0
n,18,0,0.25,0
n,35,0,0.25,0.25
n,52,0,0,0.25

et,1,beam188

ngen,17,1,1,52,17,0.25,0,0,1

E,1,2
E,2,3
E,18,19
E,19,20
E,35,36
E,36,37
E,52,53
E,53,54
E,1,18
E,2,19
E,18,35
E,19,36
E,35,52
E,36,53
E,52,1
E,53,2
E,1,35
E,53,19
E,18,36
E,36,20
E,52,36
```

```
E,36,54
E,52,2
E,2,54
E,18,2
E,2,20

Egen,8,2,all

E,17,34
E,34,51
E,51,68
E,68,17
E,51,17

/SOLU
ANTYPE,0

D,ALL,UZ,0
D,1,UX,0
D,1,UY,0
D,52,UY,0

F,26,FY,-111.2
F,43,FY,-111.2

SOLVE
FINISH

/POST1
SET,LAST

/PNUM,ELEM,1
EPLOT

PLDISP,2

PRNSOL,U,Y

ETABLE,RES1,SMISC,1
PRETAB,RES1

PRRSOL
```

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